MILITARY SPECIFICATION

RESIN, PHENOL-FORMALDEHYDE, LAMINATING

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments.

1. SCOPE

1.1 This specification covers phenol-formaldehyde resin for use as a plastic component of phenolic laminates.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standard, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

TT-P-141 —Paint, Varnish, Lacquer, and Related Materials, Methods of Inspection, Sampling and Testing.

VV-L-791 —Lubricant, Liquid Fuels, and Related Products; Methods of Inspection, Sampling and Testing.

STANDARDS

MILITARY

MIL-STD-129—Marking of Shipments.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Material shall be as specified herein.

3.1.2 *Resin.* — The resin shall be of the phenolic type, unless otherwise specified in the contract or purchase order.

3.2 The properties of the material shall be as follows:

Solids content, percent	59 to 64
Set time, tack free, seconds, at 130°	1400 to 2000
pH	7.5 to 8.0
Specific gravity at 25°/25°C.	1.144 to 1.154
Viscosity at 25°C. poises	1.25 to 1.75

3.3 Solubility.—The resin shall be completely soluble in denatured alcohol.

3.4 Form.—The resin shall be in liquid form and shall be free from foreign matter.

3.5 Workmanship.—Workmanship shall be in accordance with high grade commercial practice covering this type of material.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Lot.—Unless otherwise specified in the contract or purchase order, for purposes of sampling a lot shall consist of resin manufactured at one time, from one batch, and offered for delivery at one time.

4.2 Sampling.—Ten percent, but in no case more than 10 or less than 3 of the containers shall be selected by the inspector so as to be representative of the lot. If there are fewer than 3 containers in the lot, all the containers shall be sampled. The material in each container shall be mixed to a uniform consistency throughout and approximately 4 ounces shall be taken from each selected container. Each of these primary samples shall be placed in an airtight container and labeled so that the container from which it was taken can be identified. A composite sample of approxi-

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mately 4 ounces shall be made from equal portions of the primary samples. The composite sample shall be thoroughly mixed and placed in an airtight container labeled to show the name of the material, manufacturer, plant, contract or order number, lot number and lot size. All acceptance tests shall be made on the composite sample. However, if it becomes apparent during sampling that the lot is not uniform, the inspector may require that any primary sample be tested for compliance with the requirements of this specification. All primary samples shall be held for possible future examination should the composite sample fail to meet the requirements.

4.3 **Inspection.** - Inspection shall be at the point of delivery unless otherwise specified in the contract or order.

4.4 **Solids content, percent.**—Place a portion of the thoroughly stirred sample in a tared stoppered bottle and weigh the container and sample. Transfer 1.0 to 2.5 grams of the liquid to a tared, flat bottomed dish about 8 cm in diameter (a cover from a friction top can will be satisfactory). (A suitable piece of stout wire is weighed with the empty dish, left in it and used as a stirrer to break up any skins formed during evaporation of the volatile matter.) Spread the material over the bottom of the dish. Weigh the containers and contents again and calculate by subtracting the exact weight of the portion transferred to the weighed dish. Heat the dish and contents in an oven maintained at 105° ± 2°C. (221° ± 4°F.) for 3 hours. Any skins formed shall be broken up with the stirrer, at intervals. Cool and weigh. Reheat for 1 hour, cool and weigh. Repeat until a constant weight ± 10 mg. is obtained. From the weight of the residue in the dish and the weight of the sample taken, calculate the percent of solids.

4.5 **Set time.**—Buff a steel panel lightly using steel wool and lacquer thinner. Remove all rust and corrosion spots by sanding with No. 240 sandpaper or equal emery cloth. Rinse panel with lacquer thinner using a soft bristle brush. Wipe dry with a clean rag. Degrease panel in perchloroethylene vapor for 5 minutes. When a perchloroethylene vapor cleaning machine is not available, cold dip can also be used. In this case, the solvent must be changed frequently. Draw down a film of the material being tested to a wet film thickness of approximately .002 inch. Place the panels in a horizontal position in an oven maintained at 130° ± 1°C (266° ± 2°F.). The material should still show some tack at 1,200 seconds and be completely tack-free at 2,000 seconds.

4.6 **Determination of pH.**—Set the liquid resin to a hard mass by thorough curing at 300° ± 5° F. for 10 to 15 minutes. Grind the cured mass in a clean mortar and pestle to pass a 30 to 40 mesh sieve. Immediately after grinding, place 2 grams of the ground particles, accurately weighed, in a clean ½-ounce vial or other suitable container of chemically resistant glass. Add 10 milliliter (ml.) of freshly boiled cool distilled water and stir with a clean glass rod. Keep the glass container stoppered at all times except when pH determinations are being made. Allow the suspension of ground particles to stand for 72 hours at room temperature, stir the solution and determine the pH by means of suitable glass electrode pH meter. Determine the pH again at 96 hours and at repeated intervals of 24 hours thereafter until the difference between consecutive readings is not more than 0.05. Take the last reading as the equilibrium value for the pH of the cured resin.

4.7 **Specific gravity at 25°/25°C.**—Determine by means of a pycnometer in accordance with Method 418.3 of Specification TT–P–141.

4.8 **Viscosity at 25°C., poises.**—Determine viscosity at 25°C. in accordance with Method 30.5 (method A) of Specification VV–L–791.

4.9 **Solubility.**—Transfer an accurately weighed portion of approximately 10 gm. of the sample to a beaker and add 100 ml. of

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denatured alcohol and shake until the sample completely dissolves in the alcohol. After standing 30 minutes, examine for sediment. There shall be no settling out of homogeneous solution.

4.10 **Resubmission and retest.**—If the composite sample subjected to test, fails to pass the tests, the lot shall be rejected. The contractor shall have the option of having a partial or complete analysis made on samples taken from all or any of the containers in the lot at no expense to the Government. The contractor may then remove the defective portions of the lot and resubmit the lot for acceptance, provided complete replacement of the defective portions can be made to the satisfaction of the inspector. The resubmitted lot shall be accepted, provided that new samples, selected in accordance with 4.2, pass all the tests required by this specification.

5. PREPARATION FOR DELIVERY

5.1 **Packing.**—Unless otherwise specified, the subject material shall be delivered in standard commercial drums (55-gal. drums) so constructed as to secure acceptance by common or other carriers for safe transportation at the lowest rate to the point of delivery.

5.2 **Marking.**—Unless otherwise specified, shipping containers shall be marked with the name of the material and the quantity contained therein, as defined by the contract or purchase order under which the shipment is made, the name of the contractor, the number of the contract or purchase order, the gross weight, and the lot number. In addition, Army shipments shall be marked in accordance with Standard MIL–STD–129.

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